



The Independence Illusion

What Semiconductor Companies File — and Hope You Won't Read

A Contract-Level Diligence Reader · Nine IPOs, Read from the Exhibits Up

SEMIPPLY

Semiconductor Supply Intelligence

BUILT FROM PUBLIC SEC FILINGS · A BENCHMARK, NOT INVESTMENT ADVICE

ABOUT THIS REPORT

What it is. This report demonstrates a method: reading the agreements a company files with the U.S. Securities and Exchange Commission to surface the supply, ownership, and intellectual-property dependencies its public narrative leaves out. It is an illustration of that method — and of the depth of Semipply's semiconductor contract corpus — not a monitoring service, a rating, or a current-state assessment of the nine companies discussed.

About the data — a point-in-time snapshot. Every quotation is taken from a document exactly as filed on a specific date, most often a company's registration statement (Form S-1, F-1, or DRS) at the time of its initial public offering, or an exhibit to a later annual report. It captures each contract *as filed on that date, and no later*. Agreements quoted here may since have been amended, renewed, superseded, or terminated; ownership and control may have changed; and at least one company in this book — Aquantia — has since been acquired outright. **Nothing here should be read as the current contractual or operational status of any company.** Each chapter names the filings it draws on, and the key agreements carry their execution dates.

How to use it. The value is the lens, not the snapshot. The five questions set out in the Prologue — who controls the company, does it own or rent its technology, is its supply single-threaded, is its revenue committed or at-will, and what unwinds on a change of control — travel to any semiconductor company's own current agreements. How you apply them depends on where you sit:

- **If you invest, lend, or acquire** — run the five questions against a target's latest filings and unredacted agreements before you commit, to surface the dependencies a pitch is built to leave out.
- **If you build or run the company** — read your own contracts from the other side of the table, the way a counterparty or an acquirer already does, so you know what you have conceded before you sign the next agreement or open your next round.
- **If you sit on a board, or advise either side** — make sure management can answer these five before the board signs off on a financing, a major supply commitment, or a sale.

Where a company named here is itself of interest, pull its latest filings and any amendments before relying on anything quoted. This book shows how to read; the reading that matters is the one you do against today's documents.

A note on the quotations. All quotations are verbatim; bracketed marks such as [*] and [***] denote the issuer's own redactions, and passages set in capitals in the originals — typically warranty and limitation-of-liability language — are shown in sentence case for readability. Nothing here is investment advice, a recommendation, a credit or security rating, or a prediction, and it should not be relied upon as such. You are welcome to share this document.

FOREWORD

Every company that raises capital tells a story. It is a story of independence, of proprietary technology, of durable demand and a clear path to scale. The story is told in the prospectus summary, the investor deck, the roadshow. It is written to be read.

Underneath that story sits a second document that is written *not* to be read: the exhibits. The supply agreements, foundry contracts, license grants, and master purchase terms that a company must file with the SEC but hopes no one will actually open. They are long, redacted, and deliberately dull. And they are where the truth lives.

This short book is an experiment in reading the second document. We took nine semiconductor companies — most of them recent IPOs, all of them presented to the market as independent businesses — and read their filed contracts clause by clause. Where the SEC's reviewers had questioned the filing, we read their comment letters too, because the questions a regulator insists on are a map of what a company would rather not volunteer.

What we found was not nine separate stories. It was one structure, repeating. A company that trades as independent whose foundry, or customer, or core technology, or controlling shareholder turns out to hold a leash that the growth narrative never mentions — and that a revenue line can never reveal.

This is not a statistical study. Nine companies prove nothing about a population. What they demonstrate is a *method*: that reading the contracts a company actually signed tells you things about its resilience, its ownership, and its risk that no amount of reading its pitch ever will. The companies here are simply the ones whose exhibits we happened to read in full. We believe the pattern generalizes. We are certain the method does.

It is written for the people who write checks — investors, acquirers, lenders — and for the boards and operators of the companies themselves, who often know less about their own contractual position than the counterparty across the table does. That counterparty has signed the agreement a hundred times. You signed it once.

Read the exhibits.

— *Semipply · Semiconductor Supply Intelligence*

PROLOGUE: THE ANATOMY OF A CAPTIVE COMPANY

The independence illusion

The word that does the most work in a semiconductor prospectus is *independent*. It signals that the company controls its own destiny — its technology, its supply, its customers, its board. It is the quality investors pay a premium for, because independence is optionality, and optionality is worth money.

In all nine companies in this book, the independence is real on the cap table and qualified in the contracts. Mobileye went public as the leader in autonomous-driving vision; its filed agreements show Intel owning 80.1% of it, manufacturing the heart of its LiDAR sensor exclusively, and owning the intellectual property developed in their most strategic collaboration. Cerebras filed as the challenger to Nvidia; its contracts show a single Abu Dhabi customer, G42, that is also a major investor. Allegro trades as a Massachusetts sensor-chip company; its exhibits run almost entirely through a Japanese parent that also controls its foundry and co-owns its IP. Each of these is a real, valuable company. None of them is as independent as its story.

Five forms the leash takes

Read enough of these contracts and the dependency resolves into five recurring shapes. Most companies wear more than one.

The controlling parent. The former or majority owner whose control is written not just into the share register but into the operating agreements — and, tellingly, into the *charter*, where a waiver of the "corporate opportunity doctrine" pre-consents to the parent competing with its own subsidiary. Mobileye/Intel, Allegro/Sanken, and GlobalFoundries/Mubadala all carry this structure. The rights frequently unwind at an ownership threshold: Allegro's key agreements terminate if Sanken drops below 50%. The independence is contingent on the parent choosing not to exercise it.

The single customer. One buyer that is, functionally, the business. Cerebras and G42; Aquantia and Intel. The danger is rarely just concentration — it is that the single customer also holds every lever. Aquantia's master agreements let Intel terminate for convenience at any time and commit to no volume at all, while Aquantia carried the risk and shipped Intel's own IP under Intel's control.

The single foundry. Manufacturing single-threaded through one supplier — and often foreclosed from a second source by *contract*, not merely by cost. Everspin is barred by an exclusivity clause from taking its MRAM process to any foundry but GlobalFoundries. SkyWater's licensed process may be used "only at the Facility." SiTime buys its MEMS wafers from the one company, Bosch, that also owns the underlying patents. A second source you have contracted away is not a second source.

Rented, not owned, technology. The "proprietary" technology that turns out to be licensed — sometimes non-exclusively, sometimes revocably, sometimes with the licensee's own improvements flowing back to the licensor. MetaOptics licenses its entire metalens technology base non-exclusively from a Singapore government agency that can license the same IP to competitors and holds equity in MetaOptics besides. SkyWater licenses its core process from its former parent and assigns its improvements back. What a company *owns* versus what it *rents* is one of the sharpest distinctions in diligence, and one of the best hidden.

The change-of-control tripwire. The clause that makes independence self-terminating: rights that lapse below an ownership threshold, exclusivity that survives a sale, milestones whose failure hands the counterparty a termination right. The independence purchased at IPO can evaporate on a number in a schedule.

The asymmetry principle

Across nearly every buyer-seat agreement in this book, the same imbalance recurs: **the company commits, and the counterparty keeps its options.** The company accepts take-or-pay, non-cancellable/non-reschedulable orders, or a binding forecast. The counterparty reserves termination for convenience, non-binding forecasts, or the right to change terms "in its sole discretion." Liability caps protect the supplier; IP ownership tilts to the stronger party. A contract is a written record of who had leverage on the day it was signed — and the company being taken public is usually the party that had less.

The dependency is usually foreign

There is a geopolitical thread running through these exhibits that a domestic incorporation certificate hides. The controlling or mission-critical counterparty is foreign far more often than not: the United Arab Emirates (Cerebras's G42, GlobalFoundries' Mubadala), Japan (Allegro's Sanken, SiTime's MegaChips and Bosch of Germany), the Singapore state (MetaOptics's A*STAR), France (GlobalFoundries' substrate supplier, Soitec). This is precisely where export-control exposure (EAR/ITAR) and CFIUS risk live — and where a US-domiciled "independent" chip company can be, in operational fact, controlled from abroad. It is not visible in the revenue. It is unmistakable in the contracts.

Second-order concentration: the supplier has a supplier

Diligence that asks "who is your foundry?" usually stops at the answer. The exhibits reward the next question. GlobalFoundries anchors the supply chains of both Everspin and Aquantia in this book — and GlobalFoundries itself depends on a single French substrate supplier, via a contract inherited from AMD, and answers to an Abu Dhabi sovereign fund. Concentration does not stop at the first tier. It just gets harder to see, and almost nobody looks.

What the SEC had to drag into the open

For the companies here that drew SEC comment letters — Mobileye, GlobalFoundries, Allegro, SkyWater, SiTime — the pattern in the correspondence is as telling as the contracts. The most material dependency frequently reached the prospectus *only because the reviewer insisted*. The SEC made Mobileye add risk factors on its reliance on Intel, disclose Intel's 80.1% control, and name the customers behind 35%, 19%, and 17% of revenue that Mobileye had argued were immaterial. It made GlobalFoundries disclose its controlled-company status and its converted sovereign loans. It made Allegro identify Sanken as its controlling affiliate. The comment-letter file is, in effect, a list of the things a company would have preferred to leave in the exhibits.

The diligence lens

Strip away the specifics and the contracts answer five questions that the story is built to avoid:

1. **Who controls this company, really?** Not the cap table alone, but the rights written into its operating agreements and its charter.
2. **Does it own its technology, or rent it?** And on what terms — exclusive or not, revocable or not, improvements retained or surrendered?
3. **Is its supply single-threaded?** And if so, is a second source merely expensive, or contractually foreclosed?
4. **Is its revenue committed, or at will?** Who bears the take-or-pay, and who can walk away without penalty?
5. **What detonates on a change of control?** Which rights unwind, which exclusivities survive, which milestones hand the counterparty an exit?

Read the exhibits and you can answer all five before the first management meeting. Read only the deck and you will answer none of them — and you will learn the answers later, at a worse price.

How to read this book

Each chapter that follows is a single company, read the same way: a plain statement of the dependency, the verbatim contract language that establishes it, the questions the SEC asked where a comment-letter record exists, and a short list of questions a diligence process should put to the company. Quotations are exact; bracketed marks such as [*] and [* * *] are the issuer's own redactions, not ours. Nothing here is a prediction or a rating. It is a reading — of documents the companies themselves filed, and that anyone can check.

The companies differ in size, geography, and maturity. The structure does not. That is the point.

Contents

CHAPTER 1

Mobileye Trades as an Independent AV Leader. Its Contracts Say Intel Still Owns the Road.

CHAPTER 2

Allegro Is a "US Chipmaker." Its Foundry, Its IP, and Even Its Tax Bill Run Through a Japanese Parent.

CHAPTER 3

Cerebras's Entire Contract Book Is One Customer — And That Customer Holds a Claim on the Chips

CHAPTER 4

Aquantia Supplied Intel — on Intel's Terms. The Contracts Show a Supplier With No Leverage.

CHAPTER 5

SiTime's Foundry and Its Patent Holder Are the Same Company — and Exiting Takes Three Years

CHAPTER 6

Everspin Invented MRAM. Its Contracts Say GlobalFoundries Owns the Lock.

CHAPTER 7

SkyWater Is America's "Trusted" Merchant Foundry. Its Contracts Show What Reshoring Actually Costs.

CHAPTER 8

The Foundry Everyone Depends On Has Its Own Single Supplier — and an Abu Dhabi Owner

CHAPTER 9

MetaOptics Licenses Its Entire Technology Base — From a Singapore Government Agency.

CHAPTER 1

Mobileye Trades as an Independent AV Leader. Its Contracts Say Intel Still Owns the Road.

A read of Mobileye's 6 SEC-filed supply and IP agreements, cross-read with the SEC's questions · Semipply Semiconductor Supply Intelligence

Mobileye is the name in autonomous-driving vision — the EyeQ chips in tens of millions of cars, a 2022 IPO, a marquee independent semiconductor story. Its six filed agreements tell a more entangled version:

Mobileye's former parent Intel is its controlling shareholder, its sole photonics foundry, and the owner of the IP developed in their most strategic collaboration — while its core processor is fabbed by a single European supplier and its biggest customers have no long-term contracts at all.

Intel owns the IP that matters, and controls the terms. The Technology and Services Agreement allocates jointly developed technology in Intel's favor — *"Intel solely owns all Project IPR to Intel Core Technology"* — and the LiDAR Product Collaboration Agreement makes Intel the exclusive maker of the sensor's heart: *"Intel will exclusively manufacture the BVL2 PIC and grating and mirror wafers for the Initial Product."* The purchasing terms run one way: orders are *"non-cancellable and non-reschedulable ('NCNR'), at a minimum, during the 6 months lead time,"* and *"Intel may by notice to Mobileye change the NCNR period in its sole [discretion]."* Mobileye is locked into six months of orders it cannot cancel, on a window Intel alone can lengthen. A separate 2014 cross-license lets Intel *"terminate this Agreement ... at any time by giving notice,"* while disclaiming *"all implied warranties of merchantability, fitness for a particular purpose, and non-infringement."*

The core product depends on one other foundry. The EyeQ system-on-chip — Mobileye's flagship — traces to a long-running foundry relationship with **STMicroelectronics**, memorialized in a 2005 MOU for joint development and manufacture of EyeQ1/EyeQ2, under which *"ST will be responsible for failures due to manufacturing, testing and assembly."* So Mobileye's two most critical silicon components each run through a single supplier: photonics through Intel, EyeQ through ST.

What the SEC asked. The regulator returned to the Intel relationship again and again across five comment letters. It required Mobileye to disclose Intel's right *"to maintain 80.1% ownership,"* to add a risk factor because the charter *"waived the corporate opportunities doctrine"* — meaning Mobileye pre-agreed that Intel may take business opportunities for itself that might otherwise have belonged to Mobileye — to explain the *"\$900 million payable to Intel,"* and to identify Intel as the affiliate holding the Dividend Note — a debt owed to its parent that Mobileye repaid out of the money it raised at its IPO. It flagged Mobileye's *"reliance on Intel for silicon-photonics fabrication,"* pressed on how the LiDAR and TSA agreements *"where Intel owns most IP developed involving lidar or radar technology"* affect the business, and — separately — forced disclosure of *customer concentration*: the three largest customers at *"35%, 19%, and 17% of revenue"* with *no long-term written agreements*, which Mobileye had initially argued was immaterial. Upstream captivity to Intel and ST; downstream concentration in a handful of uncommitted customers — the SEC pulled both into the open.

The point. Cerebras is captive to G42; Allegro to Sanken; Mobileye to Intel. The pattern is the same and here it's the starkest: an "independent" IPO where the former parent holds 80.1%, makes the photonics, owns the collaboration IP, sets purchasing terms at its sole discretion, and pre-consented to its own conflicts via a corporate-opportunity waiver — while the flagship chip depends on one European fab and the revenue depends on customers who never signed up for the long term. All of it is in the exhibits, and much of it only reached the prospectus because the SEC insisted.

Questions a diligence process should ask

- What changes for Mobileye if Intel's ownership drops below the 80.1% and 20% thresholds that trigger its contractual rights?
- Given Intel owns most LiDAR/radar IP developed jointly, what does Mobileye actually own in its next-generation sensing stack?
- With EyeQ single-sourced to ST and photonics single-sourced to Intel, what is the second-source plan for either?
- How durable is revenue when the top three customers (35%/19%/17%) have no long-term agreements?

Sourced from Mobileye's SEC filings (S-1, 10-K) and SEC comment-letter correspondence. Quotations verbatim; bracketed marks denote issuer redactions.

CHAPTER 2

Allegro Is a "US Chipmaker." Its Foundry, Its IP, and Even Its Tax Bill Run Through a Japanese Parent.

A read of Allegro MicroSystems' 9 SEC-filed supply and IP agreements, cross-read with the SEC's questions · Semipply Semiconductor Supply Intelligence

Allegro MicroSystems IPO'd in 2020 as a Massachusetts sensor-chip company. Its nine filed agreements tell a quieter story than the "American semiconductor" label: **almost every one runs through two entities — Polar Semiconductor, its captive foundry, and Sanken Electric, its Japanese controlling parent.**

The parent is on every axis. When Allegro went public, Sanken still owned a majority — the SEC required it to disclose *"Sanken as an affiliate with specific percentage ownership."* That control is written into the contracts, not just the cap table. Allegro's foundry collaboration says it *"agrees not to terminate or materially alter the Addendum without consultation with and concurrence of Sanken."* Its technology IP is split — *"IP relating to magnetic sensors will be solely owned by Allegro. All other IP will be jointly owned by Sanken and Allegro."* And a royalty-sharing agreement unwinds the moment control shifts: Allegro's obligations *"shall terminate in the event that Sanken ceases to own, directly or indirectly, more than fifty percent."* If the Japanese parent's stake ever slips below half, that arrangement simply ends.

The foundry terms favor the fab. Allegro's wafers come from Polar Semiconductor under a Wafer Foundry Agreement that binds Allegro to a *"six month binding forecast"* (a take-or-pay commitment) — Allegro must take and pay for that forecasted volume even if its own demand drops — and caps liability only on the supplier's side — *"the maximum liability of PSL ... shall be limited to [XXX]"* — and can be terminated *"immediately and without liability upon written notice."* The buyer forecasts, commits, and carries the risk.

What the SEC asked. The SEC's letters were mostly about non-GAAP accounting — but the Sanken relationship kept surfacing. It made Allegro identify Sanken as the controlling affiliate, questioned a *"Polar & Sanken distribution agreement"* adjustment, and probed a *"\$9.5 million Sanken contribution"* that Allegro used to *"neutralize the cash impact"* of an IRS audit settlement. When your parent covers your tax settlement, the line between the two companies is thin — and the SEC noticed.

The point. Mobileye is captive to Intel; Cerebras to G42; Allegro to Sanken. The pattern repeats: a company that trades as independent, whose foundry, IP ownership, and even tax exposure are contractually entangled with a controlling parent — and whose key rights unwind if that parent's stake drops below 50%. It's in the exhibits, and the SEC pulled some of it into the open.

Questions a diligence process should ask

- Which agreements terminate if Sanken drops below 50% — and what's the plan if it exits?
- Is Polar a true arm's-length foundry, or a captive whose pricing and capacity Sanken influences?
- What does "jointly owned with Sanken" mean for Allegro's freedom to license or spin out non-sensor IP?

- Does the binding-forecast take-or-pay expose Allegro to inventory risk in a downturn?

*Sourced from Allegro's SEC filings (S-1/A, S-1) and SEC comment-letter correspondence. Quotations verbatim; [XXX]/[***] mark issuer redactions.*

CHAPTER 3

Cerebras's Entire Contract Book Is One Customer — And That Customer Holds a Claim on the Chips

A read of Cerebras Systems' 11 SEC-filed supply agreements · Semipply Semiconductor Supply Intelligence

Cerebras told investors that G42 — its largest shareholder — was ~85% of 2024 revenue. The contracts are blunter: **every one of the 11 supply agreements Cerebras filed is with G42 or an entity in its orbit.** There is no second customer in the record.

It all descends from one master document — a *"FRAMEWORK AGREEMENT FOR THE SUPPLY OF GOODS between G42 HOLDING US LLC and CEREBRAS SYSTEMS, INC.,"* dated 13 September 2023. When G42 became Core42 and later contracted through the Mohamed bin Zayed University of AI, the paper just notes each new order is *"on substantively the same terms as the... 'Framework Goods Agreement' entered into between Core42 Holding US LLC (previously G42 Holding US LLC) and Cerebras."* Same relationship, new letterhead.

A different kind of concentration. Single-counterparty lock-in isn't unusual for a chip company — in the same set of filings, Everspin builds its entire MRAM line on GlobalFoundries and Ambiq on TSMC. But those are *foundry* dependencies: the fab that makes your chips. Cerebras's one counterparty isn't a supplier it leans on — it's the **customer that is also its investor and its lender.** Everyone else is bound to who *makes* their chips. Cerebras is bound to who *buys* them, *funds* them, and — as the documents show — *pre-paid* them.

The prepayment, and the lever. The center of gravity is an April 2024 **Letter of Award.** G42 advances Cerebras a prepayment of *"a minimum value of three hundred million US dollars (US\$300m)"* so it can *"begin manufacturing activities... before the PO is entered into."* Then the catch — if the order isn't signed in time:

*> "any of Cerebras's rights (including ownership rights) in respect of the HPC infrastructure [***] will transfer to G42 US automatically."*

Read it twice. The customer fronted \$300M and secured an automatic transfer of ownership in the hardware if the deal slips. Buyer, investor, and fallback claimant on the IP are the same counterparty.

Washington already flagged the relationship. None of this is hypothetical. Cerebras's first IPO attempt (September 2024) was withdrawn under a CFIUS review of G42's stake, and cleared only after G42 converted to non-voting shares — a stake that keeps the economics but gives up control. A national-security review restructured the *ownership* side of a relationship whose *supply* side sits, unread, in these exhibits — and the export risk is written into the delivery terms: hardware moves US→UAE, delivery outside the US is contingent on government approval, and a failure to clear it triggers cure periods that constrain Cerebras's own right to walk.

What's dark, and what Cerebras kept. The commercial core is redacted — *"Liability Cap [***],"* warranty period [***], price [***]. But it held one line: *"Supplier may freely use and incorporate (without any compensation to G42)... any feedback provided by G42."* It kept its model-IP even as it ceded structural

control. The numbers are hidden; the leverage isn't.

Questions a diligence process should ask

- If G42's CFIUS posture shifts again, what happens to the \$300M prepayment and the automatic IP-transfer trigger?
- How much of the \$300M is already deployed to third-party vendors — i.e., unrecoverable if the deal unwinds?
- Is the framework exclusive, or is there a contractual path to a second customer?
- What is the redacted liability cap — and is it mutual, or one-way against Cerebras?

The point. A revenue footnote says "watch this customer." The contracts say Cerebras's revenue, \$300M of its balance sheet, its IP fallback, and its export clearance are all tied to a single foreign entity a US national-security review has already touched once. We pulled that from public filings alone. Imagine it on contracts that aren't public.

*Sourced from Cerebras's SEC filings (S-1, DRS/A) and the public CFIUS record. Quotations verbatim; [***] marks issuer redactions. Part of the Semipply Semiconductor Supply Intelligence research series — the deepest read of the contracts that matter.*

CHAPTER 4

Aquantia Supplied Intel — on Intel's Terms. The Contracts Show a Supplier With No Leverage.

A read of Aquantia's 4 SEC-filed supply and foundry agreements · Semipply Semiconductor Supply Intelligence

Aquantia made the 10-gigabit Ethernet PHY chips that went into enterprise networking — a real technology, later acquired by Marvell in 2019. Most of our notes look at fabless companies captive to a *supplier*. Aquantia is the mirror image: a supplier captive to a *customer*. Its filed agreements show what it's like to sell to Intel when Intel writes the contract.

Intel can leave; Aquantia can't. Three of Aquantia's four filed agreements are Master Purchase Agreements with Intel as buyer. The termination clause is one-sided in the extreme: *"Intel may terminate this Project Statement for convenience at any time by written notice to Aquantia ... Aquantia may not terminate this Project Statement for convenience."* One party holds an exit; the other is bound.

And Intel commits to nothing. The forecasts that would normally give a supplier some planning certainty are explicitly non-binding: *"Any forecasts provided by Buyer are for planning purposes only and do not constitute a Release or other commitment by Buyer."* Aquantia builds to Intel's roadmap, carries the risk, and gets no volume guarantee in return. Liability is limited to *"direct, objectively measurable damages"* with no aggregate cap identified — the protection reads as symmetric text over an asymmetric relationship.

Intel's IP sits inside Aquantia's product. The agreements embed Intel technology and control how Aquantia can use it: *"Aquantia may deliver Intel Technology ... only to those subcontractors approved in advance in writing by Intel,"* and *"the Mask Works may include trade secrets of Intel"* — the chip's own physical layout, embedding Intel's secrets. Intel controls the subcontractor list and owns IP inside the parts Aquantia ships. The customer isn't just a customer — it's a co-owner of the design and a gatekeeper on the supply chain.

The fab is GlobalFoundries. Aquantia's fourth agreement is a 2014 foundry collaboration committing it to build products on GF process technology *"as long as such GF offering meets Aquantia's product requirements."* So the picture completes: a supplier whose marquee customer can walk at will, commits to no volume, embeds its own IP, and controls subcontractors — while its manufacturing runs through the same GlobalFoundries that anchors half this series.

The point. From the buyer seat, "we supply Intel" is a logo on a pitch deck. From the contract, it's a relationship where the customer holds every lever — exit, volume, IP, subcontractors — and the supplier holds none. That asymmetry is invisible in a revenue number and obvious in the exhibits. It's also exactly the risk an acquirer should price: Aquantia's largest relationship was the one it had the least control over.

Questions a diligence process should ask

- What share of revenue rode on Intel, given Intel could terminate for convenience and committed to no volume?

- How much of Aquantia's product IP was Intel's, and what did that mean for its freedom to sell elsewhere?
- Did the GF foundry commitment leave any second-source path if GF repriced or de-prioritized?
- For any supplier: does "we sell to a marquee customer" actually mean "our biggest relationship is the one we least control"?

Sourced from Aquantia's SEC filings (DRS/A). Quotations verbatim; [] marks issuer redactions.*

CHAPTER 5

SiTime's Foundry and Its Patent Holder Are the Same Company — and Exiting Takes Three Years

A read of SiTime's 5 SEC-filed supply and IP agreements, cross-read with the SEC's questions · Semipply Semiconductor Supply Intelligence

SiTime is the leader in MEMS timing chips — the tiny resonators that replace quartz crystals. It's fabless, Santa Clara-based, and trades as a US semiconductor company. Its five filed agreements show a business built on two foreign pillars it can't easily move: **Bosch, which is both its single-source MEMS foundry and the licensor of its core MEMS IP, and MegaChips, the Japanese company that owned it and financed it into its 2019 IPO.**

One supplier is both the fab and the patents. Under the 2017 Amended and Restated Manufacturing Agreement, *"Bosch agrees to supply, and Buyer agrees to source, Buyer's [*] requirements"* — single-source MEMS wafers from Bosch. Under the separate 2018 License Agreement, Bosch grants SiTime a *sublicense* to the underlying MEMS technology, on which SiTime pays royalties and agrees to *"defend ... and hold Bosch ... harmless"* against third-party claims — if anyone sues over that technology, SiTime covers Bosch's legal bill. So the party that makes SiTime's wafers also owns the IP that makes them possible — and SiTime indemnifies that party.

The terms are supplier-friendly, and the exit is slow. Bosch's liability is hard-capped — *"in no event shall Seller's cumulative liability to Buyer ... exceed \$5,000,000 (five million U.S. dollars)"* — one of the few caps in our corpus stated as an actual number rather than [***]. And you can't leave quickly: the agreement is terminable only *"upon providing three (3) years advance written notice."* A single-source foundry that also holds your IP, whose downside is fixed at \$5M, and which takes three years to unwind, is not a supplier you switch — it's a partner you're married to.

The financing was foreign too. SiTime was owned by Japan's MegaChips, which took it public in 2019; a 2019 Integration and Purchase Agreement has SiTime supplying resonators to MegaChips. In 2023, SiTime structured an acquisition (the Master Framework Agreement with the Ningbo Aura / Aura Semiconductor entities) around *"Exclusive Licenses"* with China-territory enforcement carve-outs — a company actively managing where its IP can be enforced.

What the SEC asked. The regulator went straight at the foreign-financing web. It asked SiTime to disclose the *"current status of obligations under both the Bank of Tokyo Mitsubishi credit facility and MegaChips loan given that their maturity dates have passed,"* flagged that an exhibit *"appears to show a CEO guarantee rather than the credit increase document,"* and pressed on revenue declines tied to its *"largest end customer."* The dependencies that don't show up in a headline — Japanese debt, a personal guarantee, single-customer swings — are exactly what it made SiTime spell out.

The point. "Fabless US chip leader" is the label. The contracts say SiTime's manufacturing, its foundational IP, and its early capital all trace to foreign industrial partners it is contractually slow to leave — with a

foundry that is also its patent holder and bears only \$5M of risk. That's a resilient business *and* a concentrated one. Which it is depends on questions a revenue line never asks.

Questions a diligence process should ask

- If Bosch raised prices or de-prioritized capacity, what's the alternative — given Bosch also holds the IP and exit takes three years?
- Is the \$5M liability cap remotely adequate against a MEMS-supply failure at SiTime's current scale?
- What does the China-territory IP split in the 2023 acquisition imply about export/enforcement exposure?
- How much of SiTime's cost structure and IP freedom is set by Bosch rather than SiTime?

Sourced from SiTime's SEC filings (S-1, 8-K) and SEC comment-letter correspondence. Quotations verbatim; []/[***] mark issuer redactions.*

CHAPTER 6

Everspin Invented MRAM. Its Contracts Say GlobalFoundries Owns the Lock.

A read of Everspin's 3 SEC-filed agreements · Semipply Semiconductor Supply Intelligence

Everspin is the pioneer of MRAM — magnetoresistive memory that keeps its state without power. It designs the technology. But its three filed agreements show it doesn't control how, or whether, that technology gets built: **all three are with GlobalFoundries, and together they lock Everspin's entire MRAM line to a single foundry it funds, can't leave, and can be cut off by.**

Everspin funded the development — and transferred in its own process. The 2014 STT-MRAM Joint Development Agreement has Everspin and GF jointly building 40nm MRAM, with Everspin paying the "Project Costs." A companion Manufacturing Agreement moved Everspin's own IP onto GF's fab: *"the technology transfer of Customer's 200mm STT-MRAM technology to the GLOBALFOUNDRIES' 300mm platform."* The inventor paid to move its process onto the supplier's line.

And it can't take that process anywhere else. The exclusivity is explicit: *"Everspin shall not disclose, dispose of or license any Foreground IP to any foundry during the Exclusivity Period,"* and it covers *"any Everspin product that contains"* the jointly developed IP. There is no second foundry, by contract — not because Everspin can't find one, but because it agreed not to.

Miss a payment, lose the agreement. GF holds the leverage on both ends: *"GLOBALFOUNDRIES shall have the right to terminate this Agreement at any time if Everspin fails to pay, in full, the Project Costs,"* with a royalty-offset mechanism letting GF net down what it owes Everspin on the way out. On the manufacturing side, Everspin carries binding take-or-pay — *"a binding volume commitment on Customer's part"* — while GF's liability is capped at *"THE TOTAL PURCHASE PRICE RECEIVED BY GLOBALFOUNDRIES FROM CUSTOMER IN THE TWELVE (12) MONTHS"* preceding a claim — one year of Everspin's own payments, no more. The party that funded the technology commits the volume, carries the risk, and can be terminated for a missed check.

The point. This is the deepest lock-in in our set: an inventor whose product exists only on one foundry's line, whose IP is contractually barred from any other fab, who pre-funds the development and take-or-pay, and whose liability protection tops out at one year of its own spend. And here's the second-order twist — GlobalFoundries itself (see the companion note) depends on a single French substrate supplier and answers to an Abu Dhabi sovereign owner. Everspin's single point of failure has single points of failure of its own. That's the chain a diligence process should trace — and almost never does.

Questions a diligence process should ask

- If GF exited MRAM or repriced capacity, does Everspin have *any* contractual path to a second foundry, or does exclusivity foreclose it entirely?
- What is Everspin's remaining Project-Cost obligation, and what happens on a missed payment?

- Is the 12-month liability cap adequate against a manufacturing failure of a sole-source memory line?
- Does GF's own dependency on Soitec and Mubadala add hidden risk to Everspin's supply?

Sourced from Everspin's SEC filings (DRS/A). Quotations verbatim; [] marks issuer redactions.*

CHAPTER 7

SkyWater Is America's "Trusted" Merchant Foundry. Its Contracts Show What Reshoring Actually Costs.

A read of SkyWater Technology's 5 SEC-filed agreements, cross-read with the SEC's questions · Semipply Semiconductor Supply Intelligence

SkyWater is the rare thing US industrial policy keeps asking for: an American-owned merchant foundry, a "Trusted" supplier to the Department of Defense, expanding on public money. Its five filed agreements show the model works — and what it's built on.

It doesn't own its own process. SkyWater is a 2017 carve-out of Cypress — the filings still call it "*SkyWater Technology Foundry, Inc. (f/k/a Cypress Semiconductor (Minnesota) Inc.)*." Its core manufacturing process is licensed from Cypress, and the license is tight: the technology "*may be used by Licensee only at the Facility and may not be transferred to, or used in, any other facility,*" it's royalty-bearing, and — the striking part — SkyWater's own improvements flow back to Cypress: "*Licensee hereby irrevocably transfers, conveys, and assigns to Licensor in perpetuity all ... Licensor Improvements.*" The foundry improves the process; the former parent owns the improvement.

The national-security wrapper is contractual. A 2017 amendment exists "*to help ensure continuity of supply by Licensee to the United States Government,*" pulls in "*export-controlled ... CUI (Controlled Unclassified Information), CDI (Critical Defense Information), FOUO*" handling, and bars Cypress from assigning the agreement "*to any legal entity not organized under the laws of the United States*" without SkyWater's consent. The "Trusted Foundry" status isn't a badge — it's clauses.

The economics lean on anchor take-or-pay and public subsidy. SkyWater's 2022 Frame Agreement with **Infineon** binds the buyer to a rolling forecast with "*order commitments not less than: (i) 95% for the first Calendar Month ... (ii) 90% for the second*" — the anchor-customer commitment a capital-intensive fab needs to fund capacity. And its Florida expansion runs through a "*Technology and Economic Development Agreement ... between Osceola County, Florida and SkyWater Florida*" — public money for the fab (the BRIDG facility).

What the SEC asked. The SEC circled the same pressure point for years: how a foundry recognizes revenue. It repeatedly questioned SkyWater's "*bill and hold*" accounting — in effect, booking a sale before the chips ship — "*how the company determined control transfers during the fabrication process*" — whether "*revenue is recognized for the largest customer prior to shipment,*" the "*long-term facility expansion contract*" liabilities, and its *financial-covenant* compliance. For a subsidized fab building capacity ahead of demand, when you can book the revenue is the whole model — and the regulator kept testing it.

The point. The reshoring pitch is a domestic fab you can trust. The contracts show the trade: a foundry that licenses its process from its former parent (and hands back its own improvements), locks that process to single facilities, and funds capacity on anchor-customer take-or-pay and county money. None of that is disqualifying — it's how a merchant foundry actually gets built. But "American-owned and Trusted" is a starting point for

diligence, not the end of it.

Questions a diligence process should ask

- What happens to SkyWater's process license — and its improvements — if the Cypress relationship changes or the tech is deemed export-sensitive?
- How concentrated is capacity utilization on a few anchor take-or-pay customers, and what's the downside if one de-commits?
- How much of the growth story depends on public subsidy (Florida/BRIDG, CHIPS Act) vs. commercial demand?
- Does the "bill and hold" revenue timing the SEC flagged hold up under a demand slowdown?

*Sourced from SkyWater's SEC filings (S-1/A, 8-K) and SEC comment-letter correspondence. Quotations verbatim; [***] marks issuer redactions.*

CHAPTER 8

The Foundry Everyone Depends On Has Its Own Single Supplier — and an Abu Dhabi Owner

*A read of GlobalFoundries' 3 SEC-filed supply agreements, cross-read with the SEC's questions · Semipply
Semiconductor Supply Intelligence*

GlobalFoundries is where a large share of the fabless world's chips get made — the #3 merchant foundry, the one Everspin and Aquantia (elsewhere in this series) build their entire product lines on. Its own filed contracts show the supplier everyone leans on is itself leaning on someone — and is controlled by a Gulf sovereign fund.

GF's substrates come from one French supplier. All three of GF's filed supply agreements are with **SOITEC S.A.**, the French maker of silicon-on-insulator (SOI) wafers — the specialty substrate under GF's RF, power, and silicon-photonics processes. GF's own language is unusually candid about the dependency: *"GLOBALFOUNDRIES' operations depend upon the continuous and uninterrupted supply of Products ... GLOBALFOUNDRIES could suffer significant damage."* It commits to buy *"a minimum of [***] per calendar year"* on a *"Rolling Frozen Delivery Plan"* — take-or-pay, flowing the other direction. And the relationship is inherited: it traces to a 2003 Materials Supply Agreement between Soitec and *"Advanced Micro Devices, Inc. ('AMD')"* — the fabs GF was spun out of. GF's SOI supply chain is an AMD legacy contract with a single French counterparty.

And GF answers to Abu Dhabi. The SEC required GF to disclose it is a *"controlled company"* — controlled by Mubadala, Abu Dhabi's sovereign wealth fund. Pre-IPO, Mubadala financed it through *"shareholder loans"* the SEC questioned classifying *"within equity,"* including a *"non-interest bearing loan from shareholder,"* later *"converted ... to additional paid-in capital."* The loan was reclassified as equity, so it no longer read as debt on the books. As with other controlled companies in this series, GF *"removed ... corporate opportunity doctrine provisions"* from its charter — pre-consenting to its owner's conflicts. And its financials increasingly ride on public money: by 2024 the SEC was making GF retitle metrics because they *"include government grant proceeds."*

What the SEC asked. Beyond the standard non-IFRS cleanup, the regulator kept pulling the ownership and subsidy structure into view — the controlled-company status, the Mubadala loan mechanics, the Cayman Islands economic-substance exposure, the corporate-opportunity waiver, and the government-grant-inflated cash flow. The parts of GF's story that aren't *"leading-edge foundry"* — a Gulf owner, converted sovereign loans, grant-dependent economics — are the parts it had to spell out.

The point. Diligence on a fabless company asks *"who's your foundry?"* This note asks the next question: *who's your foundry's foundry, and who owns it?* GlobalFoundries — the anchor of a dozen supply chains — depends on one French substrate maker via an inherited AMD contract, is controlled by an Abu Dhabi sovereign fund, and leans on government grants to make the math work. Concentration doesn't stop at the fab. It just gets harder to see.

Questions a diligence process should ask

- If Soitec's SOI supply is constrained, what's GF's alternative — and which downstream products (RF, photonics, power) are exposed?
- How much of GF's control and capital-allocation sits with Mubadala, and what does the corporate-opportunity waiver permit?
- How much of GF's profitability depends on government grants vs. commercial pricing?
- For a fabless company relying on GF: is the Soitec dependency a hidden second-order single point of failure?

*Sourced from GlobalFoundries' SEC filings (F-1, 20-F) and SEC comment-letter correspondence. Quotations verbatim; [***] marks issuer redactions.*

CHAPTER 9

MetaOptics Licenses Its Entire Technology Base — From a Singapore Government Agency.

A read of MetaOptics' 5 SEC-filed IP-license agreements · Semipply Semiconductor Supply Intelligence

MetaOptics makes metalenses — flat, nanostructured optics that replace curved glass. It filed to go public as a deep-tech optics company. Its five filed agreements share one striking feature: **all five are inbound technology licenses, and all five come from the same counterparty — A*CCELERATE (Accelerate Technologies Pte Ltd), the commercialization arm of A*STAR, Singapore's government research agency.** MetaOptics does not own its core technology — it licenses it.

Every foundational license is non-exclusive and revocable. The grant language is identical across the agreements: *"A*CCELERATE hereby grants to LICENSEE a non-exclusive, non sublicensable, non-transferable, revocable for cause licence to use the Technology within the Field."* Non-exclusive means A*STAR can license the same technology to a competitor. Non-sublicensable means MetaOptics can't build a licensing business on top of it. And the licensor can pull it: if MetaOptics is *"unable to show sufficient evidence and effort to commercialise the Technology,"* A*CCELERATE *"reserves the right to terminate this Agreement with 3 months' notice."* Miss the commercialization milestones and the technology base goes away.

The government agency also holds equity. These aren't arm's-length license fees. The filings show A*CCELERATE took an *equity stake — roughly 5–6% — as the upfront licence fee*, on top of running royalties of *"1.5% of Gross Revenue" to "3% of Gross Revenue"* plus annual minimums. So the licensor is simultaneously MetaOptics' technology source, its royalty collector, and its shareholder — and audits it annually: *"A*CCELERATE may, annually ... appoint an independent auditor to examine LICENSEE's books and records."*

And MetaOptics carries all the IP risk. The licenses disclaim the one thing a licensee most needs: *"A*CCELERATE MAKES NO WARRANTY OR REPRESENTATION AS TO THE VALIDITY OR SCOPE OF THE TECHNOLOGY, OR THAT THE TECHNOLOGY WILL BE FREE FROM INFRINGEMENT."* If a third party claims the licensed metalens IP infringes, that's MetaOptics' problem — while A*CCELERATE's own liability is capped at *"any amount that LICENSEE has actually paid."* The company assumes infringement exposure on technology it neither owns nor can defend as its own.

The point. "Proprietary metalens technology" is the story. The contracts say the technology is licensed non-exclusively from a Singapore state agency that holds equity, collects royalties, audits the books, can license the same IP to rivals, can revoke on missed milestones, and provides no infringement protection. That may be a perfectly fine foundation for a young company — but it is a fundamentally different asset than *owned* IP, and the difference lives entirely in the exhibits. There's also an export wrinkle worth flagging: every license restricts the technology to *"commercial and/or civilian purposes only."*

Questions a diligence process should ask

- What does MetaOptics actually *own* versus license — and what survives if A*CCELERATE terminates for a missed commercialization milestone?
- Since the licenses are non-exclusive, could A*STAR license the same metalens technology to a competitor?
- What are the commercialization obligations (redacted in the addenda), and how close is MetaOptics to breaching them?
- Who bears a third-party infringement claim on the licensed technology, given the explicit no-warranty and paid-amount liability cap?

Sourced from MetaOptics' SEC filings (F-1/A, DRS/A). Quotations verbatim; bracketed marks denote issuer redactions.

EPILOGUE: WHAT THE NINE HAVE IN COMMON

Nine companies. Three continents of ownership, a decade of filings, technologies that run from wafer-scale AI engines to flat-optics lenses. On the surface they share almost nothing. Read their contracts and they share almost everything.

Each was presented to the market as independent. Each turned out to be tethered:

- **Mobileye**, whose former parent Intel owns 80.1% of it, manufactures the heart of its LiDAR sensor, and owns the intellectual property they developed together.
- **Allegro**, whose foundry, its IP, and even a tax settlement run through a Japanese parent whose departure would unwind its key agreements.
- **Cerebras**, whose entire contract book is one Abu Dhabi customer that is also an investor — and that holds a claim on the chips themselves.
- **Aquantia**, a supplier whose largest customer could terminate at will, committed to no volume, and embedded its own IP inside the parts it bought.
- **SiTime**, whose sole foundry is also the owner of its core patents, with a three-year exit and a five-million-dollar ceiling on the supplier's risk.
- **Everspin**, the inventor of its category, contractually barred from taking its own process to any foundry but one.
- **SkyWater**, America's "trusted" domestic foundry, licensing its process from its former parent and assigning its own improvements back.
- **GlobalFoundries**, the foundry a dozen others depend on — itself dependent on a single French substrate supplier, and controlled by a sovereign fund.
- **MetaOptics**, an "independent" company whose entire technology base is licensed, non-exclusively, from a government agency that also holds its equity.

Read as a set, the lesson is not that any of these is a bad company. Several are excellent. The lesson is that **independence is a claim, and the claim is testable** — and the test is not in the growth chart or the management call. It is in the exhibits, where the counterparty's leverage is written down, in language the counterparty chose.

The market prices the story. The story is written to be read, and it is read. The contract is written to be filed, and it is filed — and then not read. That gap is the whole opportunity. The party across the table from every one of these companies has signed its version of the agreement hundreds of times and knows precisely what it holds. The company signed once. The investor, the acquirer, and the lender arrive later still, and most often read the deck.

We built Semipply to close that gap: to read the semiconductor contracts that matter — from your side of the table — and to benchmark what a company signed against what its peers actually agreed to, drawn from real, unredacted SEC filings. Not a score. Not a prediction. A reading you can check, clause by clause, against the source.

Before the next check clears, the next term sheet is signed, or the next supply agreement is renewed:

Read the exhibits.

Semipply · Semiconductor Supply Intelligence · legalforensics.ai